

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining
COURSE CODE : CSA1610

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction , for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

<i>Criteria</i>	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I TEJASWINI.M(192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the annual premium increase rate (x_i) and the number of claims per 100 policyholders (y_i) in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a positive or inverse relationship.

Before computing covariance, calculate the mean of x and y .

Premium Increase % (x_i)	Claims per 100 Policyholders (y_i)
3.2	15
4.0	18
5.5	25
4.8	20

SOLUTION:

To determine whether the premium increase rate and claim frequency have a positive or inverse relationship using covariance.

Given Data

Premium Increase (x)	Claims (y)
3.2	15
4.0	18
5.5	25
4.8	20

Step 1: Calculate Mean

$$\bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4} = 4.375$$
$$\bar{y} = \frac{15 + 18 + 25 + 20}{4} = 19.5$$

Step 2: Covariance Table

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

Sum = 12.25

Step 3: Covariance

$$Cov(x, y) = \frac{12.25}{4} = 3.0625$$

Result

Covariance = 3.0625

Since the covariance is positive, premium increase and claim frequency have a positive relationship.

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing patient blood pressure readings (systolic) to identify trends. Due to minor fluctuations, the analyst performs local smoothing using equal-frequency (equi-depth) binning.

Sample Data Set (Systolic BP readings):

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

SOLUTION:

Given Data

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Total values = 27

Divide into 3 equal-frequency bins (9 values each).

Bin 1

110, 112, 115, 115, 118, 120, 120, 122, 124

Mean

$$= \frac{1056}{9} = 117.33$$

Smoothed Values

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33

Bin 2

124, 126, 126, 126, 126, 130, 135, 135, 138

Mean

$$= \frac{1166}{9} = 129.56$$

Smoothed Values

129.56 repeated 9 times.

Bin 3

138, 138, 140, 145, 150, 152, 160, 180

Mean

$$= \frac{1203}{8} = 150.38$$

Smoothed Values
150.38 repeated 8 times.

Result

Equal-frequency binning smooths the blood pressure readings by replacing values with the bin mean, reducing minor fluctuations.

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates Body Mass Index (BMI) of 18 clients for risk classification. To simplify analysis, they smooth the data by replacing values with closest bin boundaries using equal-width partitioning.

Sample Data Set (Sorted BMI):

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

SOLUTION:

Given BMI Data

18.5,19.2,21.8,24.9,25.5,26.2,26.4,27.8,29.2,30.2,30.4,31.9,32.4,33.1,33.6,34.7,38.2,39.5

Step 1

Minimum = 18.5

Maximum = 39.5

Range

$$39.5 - 18.5 = 21$$

Number of bins = 3

Bin Width

$$= \frac{21}{3} = 7$$

Bins

Bin 1: 18.5 – 25.5

18.5,19.2,21.8,24.9,25.5

Boundary Values

18.5,18.5,18.5,25.5,25.5

Bin 2: 25.5 – 32.5

26.2,26.4,27.8,29.2,30.2,30.4,31.9,32.4

Boundary Values

25.5,25.5,25.5,32.5,32.5,32.5,32.5,32.5

Bin 3: 32.5 – 39.5

33.1,33.6,34.7,38.2,39.5

Boundary Values

32.5,32.5,32.5,39.5,39.5

Result

Equal-width binning classifies BMI values into risk groups using fixed-width intervals.

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the cost of medical consumables (in ₹).

To reduce data for mining purposes, they discretize the sorted price records using bin medians.

Sample Data Set (Costs in ₹):

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

SOLUTION:

Given Data

120,140,145,145,148,150,150,150,155,155,155,155,160,160,165,170,170,175,175,175,175,180,190,200,210,230,300

Bin 1

120,140,145,145,148,150,150,150,155

Median = **148**

New Values

148,148,148,148,148,148,148,148,148

Bin 2

155,155,155,160,160,165,170,170,175

Median = **160**

New Values

160 repeated 9 times.

Bin 3

175,175,175,180,190,200,210,230,300

Median = **190**

New Values

190 repeated 9 times.

Result

Replacing each value with the bin median reduces data size while preserving the central tendency.

5. Standardization of Patient Age Data (5 Marks)

Given the following ages of insured individuals:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

- (a) Compute the mean absolute deviation (MAD) of age.
- (b) Compute the z-score for the first four observations.

SOLUTION:

Given Ages

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Step 1: Mean

$$\bar{x} = \frac{330}{10} = 33$$

Step 2: Mean Absolute Deviation (MAD)

$$MAD = \frac{88}{10} = 8.8$$

Step 3: Standard Deviation

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum (x - \bar{x})^2}{10}} \\ &= \sqrt{115} = 10.72\end{aligned}$$

Step 4: Z-score

Formula

$$z = \frac{x - \bar{x}}{\sigma}$$

Age Z-score

18	-1.40
22	-1.03
25	-0.75
42	0.84

Result

- Mean = 33

- **MAD = 8.8**
- **Z-scores**
 - $18 \rightarrow -1.40$
 - $22 \rightarrow -1.03$
 - $25 \rightarrow -0.75$
 - $42 \rightarrow 0.84$